Loco unt 30.5.24 1.01 MBJ

**SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI**
**Mechanical**
**IA/IB/IC Schedule of 3Phase Electric Locomotive [Inspection]**

पश्चिम रेलवे Western Railway

लोको संख्या Loco No 32649 क्लास Class G9H शिड्यूलमें लेने का दिनांक Date taken under Sch IA

| | आगमनIncoming / शिड्यूलSchedule | | अंतिमनिरीक्षणFinal Inspection | |
|---|---|---|---|---|
| | अपर डैक Upper Deck | अंडरट्रकUnder Truck | अपर डैक Upper Deck | अंडर ट्रक Under Truck |
| Date/ Shift | 8/7-25/3Rd | 8/7 25/3rd | | |
| Name of Technician | RAVINDRA + Di | [illegible] | | |
| Designation/ Token No. | DINESH / Tech | M.1 | | |
| Signature | [illegible] | [illegible] | | |

**Customer Feed Back/Bookings** (Record feedback given by Loco pilot)

| Sr. No. | Customer Feed Back/Bookings | Action Taken | Name of TCN / Attended By | Remarks, if any & Sign. of JE/SSE |
|---|---|---|---|---|
| 1 | Panto 2 Isolat | | | |
| 2 | Cab 2 ALP Side Send pipe Brokan | fitted | | |
| 3 | All Send Not working | attend | | |
| 4 | Both Side MR & BC Pipe missing | | | |
| 5 | CP2 Inter Colan not working | | | |
| 6 | LR/Sand conver 1x4 dummi missing | fitted | | |
| 7 | LL Side Sand con 1 No dumi miss | fitted. | | |
| 8 | CP 2 Fem Protonction galu damage | attend | | |
| 9 | CAB 2 wiper water system not work | | | |

**Inspection Wing Mechanical: Check and record Following**

| | CPA Loading Time (Pressure 0 to 8.0 kg/cm$^2$) | Safety Valve Blowing | Pantograph-1 | | Pantograph-2 | | Compressor Loading Time (Pressure 8 to 9.5 kg/cm$^2$) | | CP Safety Valve Blowing | Unloaded valve |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Raising | Lowering | Raising | Lowering | Comp-1 | Comp-2 | | |
| Std. Value | 8.0 Minutes | 8.5± 0.25 kg/cm$^2$ | 6 to 10 Second | Less than 15 Second | 6 to 10 Second | Less than 15 Second | 120 Second | 120 Second | 11.5± 0.35 kg/cm$^2$ | Operation time Approx. 12 Sec. |
| GI | | ok | 6.90 | 7.51 | 6.82 | 5.17 | 26.0s | 26.5s. | ok. | 5. |
| Final | | ok | 7.10 | 9.58 | 7.60 | 7.18 | 28.0 | 30 sec | ok | 6. |

| | MR safety Valve pressure | | Duplex check valve setting/Brake Pipe Pressure by Auto Brake (A9) | | Feed Pipe Pressure | | Brake Cylinder Pressure by Direct Brake (SA9) | | Horn Sound | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Blowing | Closing | | | | | | | Std. - Good | | | |
| Std. Value | 10.5± 0.2 kg/cm$^2$ | 9.5–9.8 kg/cm$^2$ | 5.0 ± 0.1 kg/cm$^2$ | | 6.0 ± 0.2 kg/cm$^2$ | | 3.5±0.1 kg/cm$^2$ | | Cab-1 | | Cab-2 | |
| | | | Cab-1 | Cab-2 | Cab-1 | Cab-2 | Cab-1 | Cab-2 | LP | ALP | LP | ALP |
| GI | ok | ok | 5.0 | 5.0 | 6.1 | 6.2 | 3.8 | 3.8 | G | G. | G | G. |
| Final | OK | OK | 5.0 | 5.0 | 6.1 | 6.2 | 3.6 | 3.8 | G | G | G | G |

| | Computerized Control Brake (CCB) System if applicable | | Air Flow | Air Dryer | | ADC | MRC Time By both CP 0-10kg/cm$^2$ |
|---|---|---|---|---|---|---|---|
| Std. Val. | | Alerter and Penalty | Working | Working, Tower to change every minute (FTIL, SIL) & In every two minutes (KBIL) | Humidity Indicator Colour Std-Blue | Working | 3.5 Minutes (Max.) For 1750 LPM or [8.5 Minute for each 1450 LPM CP] |
| | Passed successfully | Working | | | | | |
| GI | | | W. | W | BLUE | W | 3.40 Sec. |
| Final | | | W | TDENP | blue | W | 3.30. Sec |

- All gear case oil to change is viscosity very height as per lab report

Signature of QAG JE/SSE

DP [signature]

| | CPA Pressure build up time 8.5 $kg/cm^2$ | BP drop in 5 Minute | FP drop in 5 Minute | MR drop in 15 Minute | Press BPPB to apply or release parking brake. Check Pressure in Parking Brake gauge | | VCB Pressure switch setting |
|---|---|---|---|---|---|---|---|
| | | | | | 6.0± 0.15 $kg/cm^2$ | | |
| Std. Value | 60 Sec. Max. | Max. drop 0.15 $kg/cm^2$ | Max. drop 0.70 $kg/cm^2$ | Max. drop 1.0 $kg/cm^2$ | Cab-1 | Cab-2 | Opens 4.5± 0.15 $kg/cm^2$, Closes 5.5± 0.15 $kg/cm^2$ |
| GI | 58.15 sec | 0.0 | 0.0 | 0.5 | | | 5.1 |
| Final | 60 sec | 0.0 | 0.0 | 0.6 | | | 5.1 |

| | ACP on 4mm test plate | | Auto brake controller Position | | BP ($kg/cm^2$) | | BC ($kg/cm^2$) | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Cab-1 | Cab-2 | | Cab-1 | Cab-2 |
| Std Values | Working | | Run | 5.0 ± 0.10 | 5.0 | 5.1 | 0.0 | 0.0 | 0.0 |
| | | | Initial Application | 4.6 ± 0.10 | 4.6 | 4.6 | 0.40 ± 0.10 | 0.4 | 0.4 |
| | Cab-1 | Cab-2 | Full Service | 3.35 ± 0.20 | 3.5 | 3.4 | 2.50 ± 0.10 | 2.5 | 2.5 |
| | | | Emergency | Below 0.30 | 0.0 | 0.0 | 2.50 ± 0.10 | 2.5 | 2.5 |
| Gen. Insp. | W | W. | | | | | | | |
| Final | | | | | | | | | |

| | Loco pilot cabin condition | | Wiper Operation | | Look out Glass | | Fire Extinguisher | Sand box | Sander Operation | | Cleaning of Loco |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Cab-1 | Cab-2 | Cab-1 | Cab-2 | Cab-1 | Cab-2 | | | Cab-1 | Cab-2 | |
| Std. Values | Good | Good | Working | Working | Clear | Clear | Sealed | Full | Working | | Cleaned |
| Gen. Insp. | D | D | ok | W. | C | D | Sealed | - | | | |
| Final | C | C | W | W | C | C | Sealed | | | | Cleaned. |

| क्र.सं. SN | कार्यवाही कार्य /निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule) | कार्यवाही की गयी Action Taken | टीमीएन का नाम Name of TCN | हस्ताक्षर/ टिप्पणी Sign/ Remarks |
|---|---|---|---|---|
| **(A) Pneumatic System (When Loco is energised check, attend and record)** | | | | |
| 1 | Check correctness of reading of BP, FP, MR, BC, AFI & RS gauges. | ok | | Checked |
| 2 | Check on dummy of BP/FP pipe. | NA | | Checked |
| 3 | Check the working of PTDC | NA | | Checked |
| 4 | Check the proper functioning of all the gauges of the driver desk<br>BP: (i) Normal (ii) with 7.5 mm hole BP should not drop below 4 $kg/cm^2$ in 60 second.<br>FP: (i) Normal (ii) with 5.5 mm hole FP should not drop below 4.5 $kg/cm^2$ in 60 second.<br>MR: (i) Mainline MR leakage without application of SA-9 Direct Brake<br>(ii) Mainline MR leakage with application of SA-9 Direct Brake | FP:- 6.0<br>BP:- 4.6 | | |
| 5 | (RS drop) Panto line air leakage 0.70 $kg/cm^2$ in 5 minutes. | ok | | |
| 6 | Check working of Panto selection switch. | ok | | |
| 7 | Position of isolating cock on pneumatic panel. (70,74,136 in open condition & 47 in close condition.) | ok | | |
| 8 | Drain moisture from MR and Inter Cooler. | OK | | |
| 9 | Check functioning of air dryer for purging, cycle time and change over with MR pressure of 8.0 $kg/cm^2$. | OK | | [illegible] 25/5/24 |
| 10 | Check Position of the Air dryer cock (13,14 in service & 187 in isolated) | OK | | |
| 11 | **Check air dryer charging when compressor is in working. (FTIL -120 sec/SIL-180 sec)** | OK | | |
| 12 | Check and attend leakage of air dryer, compressor valves, Pneumatic valves, Pneumatic panels & pipelines. | | | |

- BC. pr. 4.0 kg to adjust (cab 1)
FP. pr. 6.1 to adjust. (cab 1)

Signature of QAG JE/SSE

| क्र.सं. SN | कार्यवाही कार्य /विवरण निरीक्षण का /(शिड्यूल) Detail of Work/ Inspection (Schedule) | कार्यवाही की गयी Action Taken | टीसीएन का नाम Name of TCN | हस्ताक्षर/ टिप्पणी Sign/ Remarks |
|---|---|---|---|---|
| **B** | **BRAKE CONTROLLER (AUTO AND DIRECT)** | | | |
| 1 | Check the functioning of driver's direct air brake & automatictrain brake. | checked ok | 25/8/24 | |
| 2 | Check the free movement of the brake controller. | checked ok | | |
| 3 | Perform repeated application and release of brakes with both auto and direct brake controller from both cabs and ensure that BP & BC pressure build up & drop is proper on the gauge. Ensure by physically going to individual main and parking brake cylinder that their operation is smooth and without air leakage. | checked ok | | |
| **C** | **CARBODY FILTER:** Check condition and clean car body filter and refit | clean & filled | 26-08-24 | |
| 1 | Thoroughly clean all roof and side body filters with water jet. | | | |
| 2 | Check tightness of Roof joint clamp and Roof top bolt. | | | |
| 3 | Remove Louvers& Fly screen and clean it. Repair if found defective. | | | |
| 4 | Remove the tread plate and clean the drainage channels.Check and adjust, if necessary, the door-operating rod. **(IC Sch)** | | | |
| **H** | **LOOKOUT GLASS & SIDE WINDOWS** | | | |
| 1 | Clean any built up dirt and debris from the channel below the cab-sliding window. Clear the sliding window channel drains hole in the door opening. | checked ok | | |
| 2 | Clean look out glass with detergent/water solution. | checked ok | | |
| **I** | **WASHER / WIPER** | | 25/8/24 | |
| 1 | Check the operation of windscreen wipers using the manual handles. Rectify any fault found. | checked ok | | |
| 2 | Check the condition of windscreen wiper blades. Replace if required. | checked ok | | |
| 3 | Check the wiper and parallel arm assembly for wear anddamage. Replace the arm or any part if there is excessive wear or damage. (IC Sch) | checked ok | | |
| 4 | Check the water level in the reservoir at No.1 and No.-2 end.Top up if required. Check wipers for proper operation | checked ok | | |
| 5 | Clean the washer nozzle jet. (IC Sch) | checked ok | | |
| 6 | Check for damage of air and water pipes | | | |
| 7 | Examine all components for wear, looseness scoring andcracks, renew any defective components. | checked ok | | |
| 8 | Check seat-mounting points for integrity and ensure fixing or fully tightened. | checked ok | | |
| J | Key interlocking: Check all keys are present & the complete system is operational | checked ok | | |
| **K** | **Fire Extinguisher** | | | |
| 1 | Ensure availability of fire extinguishers and ensure that it is not expired. | checked ok | 25/8/24 | |
| 2 | Check the condition of CO2 gas cylinder, its regulator and pressure gauge in both cabs. Check the CO2 gas pressures. Refill if necessary | checked ok | | |
| 3 | Clean pipeline by compressed air and check that there should not be any leakage. | checked ok | | |
| 4 | CO2 Gas pressure and date: Cab-1 82 kg 30-12-24 / Cab-2 81 kg. 30-12-24 | | | |
| | | | | |

Signature of QAG JE/SSE

| क्र.सं. SN | कार्यवाही कार्य /निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (Schedule) | कार्यवाही की गयी Action Taken | टीसीएन का नाम Name of TCN | हस्ताक्षर/ टिप्पणी Sign/ Remarks |
|---|---|---|---|---|
| 5 | Clean fire detection unit pipeline air intake holes with soft non-metallic brush & vacuum cleaner. | | | |
| L | **OIL LEAKAGE IN MACHINE ROOM / CLEANING OF MACHINE ROOM.** | | | |
| 1 | Visually inspect the machine room for any signs oil leakage from traction converter, transformer, D.C. Link, and series resonant circuit capacitors and other electrical equipment and carefully clean the same. **(Caution : Electric supply room)** | Checked ok | | |
| 2 | Clean the complete machine room with the air pressurizedsupply and vacuum cleaner to remove all the dust particles. | | [illegible] 25/5/24 | |
| M | **SAND GEAR SYSTEM** | | | |
| | a) Check Sand box, cover gasket, Ejector and replace gasket if necessary. | checked ok | | |
| | b) Check all sander pipe for alignment and check fasteners also. | checked ok | | |
| | c) Check and ensure tightness of Sand box foundation bolt. | checked ok | | |
| | d) Ensure sander pipe height 60mm above rail level | checked ok | | |
| | e) Refill sand in all sand boxes | | | |
| N | **OIL COOLING UNIT CASING WITH RADIATOR** | | | |
| 1 | Remove all dust, dirt and debris from the radiator chamber via the machine room access cover. (By using vacuum cleaner) | | | |
| 2 | Visually check the conservator tank along with its gauge glassfor any sign of crack / damage / oil leakage. | | | |
| 3 | Measure air delivery before and after cleaning of radiator | | Ramesh | [illegible] 26/5/24 |
| 4 | Measure vibrations of oil cooling casing using vibration analyzer and take appropriate action | done. | | |
| 5 | Visually check the casing and support arrangement for oilpumps on the casing for any sign of crack of the damage | | Lalit | |
| 6 | Visually check the oil cooler radiator for any oil leakage from VPTFP and VPSR and externaldamage from top to bottom. | | Dhanraj | |
| 7 | **IC Sch: a)** Clean radiator with hot water jet and check that the radiator is not blocked. | | | |
| 8 | b) Tighten the front foundation bolts of casing mounted on to theradiator and check intactness of jail and Z-clamp welding crack. | | | |

**Check and put (√) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action taken in the Non-Schedule work table.**

**UPPER DECK [Inspect, Attend and Record]**

| SN | Description | CB | Remarks |
|---|---|---|---|
| 1. | Loose, missing and broken components/parts. | √ | |
| 2. | Clamping and condition of its welding/fastener. | √ | |
| 3. | Condition and working of all pressure gauges. | √ | |
| 4. | Working of all brake valves. | √ | |
| 5. | Leakage of air (MR) | √ | |
| 6. | Gauges, securing of loco pilot cabin. | √ | |
| 7. | Operation of Wipers. | √ | |
| 8. | Co-Action working of A9 Auto brake. | √ | |
| 9. | Fire Extinguishers (prescribed Nos.) | √ | |

**Signature of QAG JE/SSE**

| | | | |
|---|---|---|---|
| 10. | Check tightness of compressor foundation bolt & abnormal sound. | ✓ | |
| 11. | check slackness of compressor fan and no vibration in comp. | ✓ | |
| 12. | Check tightness of air dryer foundation bolt. | ✓ | |
| 13. | Ensure Air dryer indicator blue colour | ✓ | |
| 14. | Ensure clamping of all pipe to avoid vibration and rubbing. | ✓ | |
| 15. | Ensure fitment of all split pins | ✓ | |
| 16. | tightness of all bolts of Loco Roof Check | ✓ | |
| 17. | Check window of both cab, shutter window door & corridor doors | ✓ | |

**Check and put (√) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action taken in the Non-Schedule work table.**

**UNDER TRUCK [Inspect, Attend and Record]**

| क्र.सं. SN | विवरणDescription | CB | टिप्पणीRemarks |
|---|---|---|---|
| 1. | Loose, missing and broken components/parts. | ✓ | |
| 2. | Clamping and condition of its welding/fastener. | ✓ | |
| 3. | Air leakage in bogie. | ✓ | |
| 4. | Overheated parts / Abnormal wear in parts. | ✓ | |
| 5. | Check and Ensure Brake block and Wheel clearance >10mm | ✓ | |
| 6. | | | |

---

**(Remarks if any & sign):**

**Signature and Name of Technician for Inspection Incoming with date & shift:**

**Signature and Name of JE/SSE for Inspection Incoming with date & shift:**

**Signature and Name of Technician for Inspection Final with date & shift:**

**Signature and Name of JE/SSE for Inspection Final with date & shift:**

**अन्य किसी भी जानकारी Any Other Information:**

***Signature of QAG SSE Incharge (Inspection)***

**Signature of QAG JE/SSE**

## SCHEDULE PROGRESS CHART

| SN | Date | Shift | QC | Work Details | Name of JE/SSE | Sign of JE/SSE | Remarks |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |
| 4 | | | | | | | |
| 5 | | | | | | | |
| 6 | | | | | | | |
| 7 | | | | | | | |
| 8 | | | | | | | |
| 9 | | | | | | | |
| 10 | | | | | | | |
| 11 | | | | | | | |
| 12 | | | | | | | |
| 13 | | | | | | | |
| 14 | | | | | | | |
| 15 | | | | | | | |
| 16 | | | | | | | |
| 17 | | | | | | | |
| 18 | | | | | | | |
| 19 | | | | | | | |
| 20 | | | | | | | |
| 21 | | | | | | | |
| 22 | | | | | | | |
| 23 | | | | | | | |

**Signature of QAG JE/SSE**

**SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI**

**IA/IB/IC SCHEDULE OF 3 PHASE ELECTRIC LOCOMOTIVE [UNDERFRAME]**

पश्चिम रेलवे
Western Railway

| लोको संख्या Loco No | 32649 | क्लास Class | IA (G9H) | शिड्यूल में लेने का दिनांक Date taken under Sch | IA |
|---|---|---|---|---|---|

**UNDERFRAME IA/IB/IC SCHEDULE & RECORDS**

| A-Check and record Hoot Wear, Flange Wear & Tread Wear (Record Wheel Diameter in Schedule or After TT) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Wheel No | Wheel Tread dia | Root Wear | Flange Wear | Tread Wear | Wheel No | Wheel Tread dia | Root Wear | Flange Wear | Tread Wear |
| 1 | 1029 | 2 | 1 | 1 | 7 | 1034.9 | 1 | 0 | 1 |
| 2 | 1028.5 | 1 | 0 | 1 | 8 | 1034.5 | 2 | 1.5 | 1 |
| 3 | 1028.2 | 3 | 1.5 | 1 | 9 | 1035.00 | 1.5 | 0 | 1.5 |
| 4 | 1028.00 | 1 | 1 | 1 | 10 | 1035.3 | 3 | 1.5 | 1.5 |
| 5 | 1027.7 | 1 | 0 | 1 | 11 | 1035.2 | 2 | 1 | 1 |
| 6 | 1027.00 | 1.5 | 1 | 1 | 12 | 1035.0 | 2 | 1.5 | 1 |

25/5/24

B- Check and Record in IC Sch Vertical &lateral clearance (Primary & Secondary) as per TC/0082 (REV-1) dated 17/09/2015

| Standard Value | Primary vertical clearance (Cab-1 &2 Bogie) Actual Value Axle box & Bogie frame | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| 27 to 35 mm | | | | | | | | | | | | |

| Standard Value | Primary lateral clearance (Cab-1&2 Bogie) Actual Value Axle box & Bogie frame | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| 15 to 22 mm | | | | | | | | | | | | |

NA

| Standard Value | Secondary lateral clearance (Cab-1 Bogie) Actual Value Bogie frame & under frame | | | |
|---|---|---|---|---|
| | CAB-2 LP Side | CAB-1 ALP Side | CAB-2 LP Side | CAB-1 ALP Side |
| 32 to 60 mm | | | | |

| Standard Value | Secondary lateral clearance (Cab-1 Bogie) Actual Value Bogie frame & under frame | | | |
|---|---|---|---|---|
| | CAB-2 LP Side | CAB-1 ALP Side | CAB-2 LP Side | CAB-1 ALP Side |
| 45 to 55 mm | | | | |

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DE | 37.1 | 37.5 | 36.9 | 37.2 | 37.6 | 38.1 | 37.7 | 37.3 | 38.1 | 38.3 | 37.9 | 37.8 |
| NDE | 35.1 | 35.3 | 35.4 | 36.7 | 36.5 | 36.0 | 35.5 | 35.1 | 36.3 | 36.0 | 35.7 | 35.3 |

| MSU | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| gear case | 40 | 40.2 | 41 | 40.3 | 42 | 41.4 |

क्यूएजी जेई/एसएसई QAG-JE/SSE

25/5/24

**C- Check and record wheel gauge below Parameters: (IC Schedule) WHEEL GAUGE [Permissible limit 1596 mm, (159610.5 mm for RB/New loco)]**

| Axle No. | A | B | C | Average | Remarks |
|---|---|---|---|---|---|
| 1 | | | | | |
| 2 | | | | | |
| 3 | | | | | |
| 4 | | NA | | | |
| 5 | | | | | |
| 6 | | | | | |

Check and record below Parameters in Schedule or if TT is done:

| Axle No. | Standard (mm) | | Cab-1 | | Cab-2 | |
|---|---|---|---|---|---|---|
| Location | Max | Min | Rl | Ll | R2 | L2 |
| Buffer Length | | | 635 | 636 | 635 | 636 |
| Buffer Height | | | 1075 | 1070 | 1075 | 1070 |
| Rail guard height | | | 107 | 107 | 107 | 107 |
| CBC Height | | | 1065 | | 1060 | |

| | | Cab-1 | Cab-2 |
|---|---|---|---|
| STRIKER WEAR PLATE | New 8 mm, Condemn 2 mm | | |
| SHANK WEAR PLATE | New 6 mm, Condemn 1 mm | | |

**E- Measure D SHAKLE and write Coupling No.**

CAB-1 OK CAB-2 OK

Check and Record Gear case Oil Level:

| TM | OIL FOUND | OIL TOP UP | TM | OIL FOUND | OIL TOP UP |
|---|---|---|---|---|---|
| 1 | OK | full | 4 | OK | full |
| 2 | OK | full | 5 | OK | full |
| 3 | OK | full | 6 | OK | full |

| | Hand Brake Operation | CB Coupler Operation & Condition | | Transition Screw Coupling Operation & Condition | | CBC Lock Pin | | HP/FF Pipe | |
|---|---|---|---|---|---|---|---|---|---|
| Std. Value | Working | Normal | | Normal | | Intact | | Intact | |
| | | Cab-1 | Cab-2 | Cab-1 | Cab-2 | Cab-1 | Cab-2 | Cab-1 | Cab-2 |
| Gen. Insp. | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | |
| Final Insp. | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | |

| Brake Piston Travel : | | | | | | | Brake Shoe release | Standard 3/9/10 mm |
|---|---|---|---|---|---|---|---|---|
| Class of Loco | Standard value | Observed (mm) | | | | | | |
| WAG7/WAP7 | 107 to 117 mm | Location- | 1 | 2 | 3 | 4 | | |
| WAG9 | | Location- | 107 | 107 | 107 | 107 | | |

25/5/24

क्यूएजी जेई/एसएसई QAG-JE/SSE

| Sr. No. | Detail of Work/ Inspection | Name Technician | Sign/Remark |
|---|---|---|---|
| F | **DRAFT GEAR AND COUPLERS** | | |
| | 1. Liberally apply general-purpose grease to the face of the centre buffer coupler at the end of the Locomotive | Checked | |
| | 2. Check the intactness of the CBC bottom support plate bolts | Checked | |
| | 3. Check the knuckles against crack by DPT (IC Sch) | Checked | |
| | 4. Check knuckle by gauge no. 1&2 (130-135 mm by go no go gauge) (IC Sch) | Checked | |
| | 5. Check knuckle nose wear by gauge no 3 go no go gauge (IC Sch) | Checked | |
| | 6. Check intactness and functioning of CBC operating handle | Checked | |
| | 7. Check visually the striker casting of CBC assembly | Checked | |
| | 8. Check for working of CBC. Lock and unlock it, locking arrangement, working of TC | Checked | |
| | 9. Check washer lock at knuckle pin bottom | Checked | |
| | 10. Check gap between Striker and Draft gear for draft gear slackness | Checked | |
| G | **BUFFERS** | | |
| | 1. Check buffers for any crack visually of the flange and intactness of foundation bolts, check nuts and split pins (4 nos. each). | Check ok | |
| | 2. Liberally apply general purpose grease to the face of the buffers at the ends of the Locomotive. Also smear grease on the buffer head plunger | Check ok | |
| | 3. Conduct RDPT of Buffers to detect crack flaws. (IC Sch | Check ok | 25/5/24 |
| | 4. Check and measure the buffer height and buffer length. (IC Sch) | Check ok | |
| H | **SUSPENSION** | | |
| | 1. Examine the primary and secondary springs for signs of damage, cracks and broken end visually with torch light and tapping by a hammer & replace, if necessary | Checked | |
| | 2. Examine different types of dampers for oil leakage and intactness of fasteners and end fittings | Checked | |
| | 3. Proper torque of all fasteners related to suspension arrangement and mounting of equipment in under frame (IC Sch) | Checked | |
| | 4. Check the condition of various Spheris block in the primary and secondary suspension arrangement and of various equipment mounted in the under frame like axle guide link, traction motor and its support arm, gear case support arm for their healthiness (it should not be crack) and replace, if necessary | Checked | |
| | 5. RDPT of all TM nose and its support arm. (IC Sch) | Checked | |
| I | **TRACTION BAR** | Checked | |
| | 1. Check the well being and presence of the safety sling | Check ok | 25/5/24 |
| | 2. Check the push/pull rod pivot heads for damage | Check ok | |
| | 3. Examine the traction link assembly for signs of damage. Any defective item should be renewed | Check ok | |

क्यूएजी जेई/एसएसई QAG-JE/SSE

| Sr. No. | Detail of Work/ Inspection | Name Technician | Sign/Remark |
|---|---|---|---|
| | 4. Check the fastener at the traction ends and those retaining the traction rod for security. If a fastener is loose, it must be renewed and tightened to the specified torque. Ensure intactness of the locking plate. SM1-241 | Check ok | |
| | 5. RDPT of traction bar at end (flange) portion & V-Ring | Check ok | |
| | 6. Housing flange. 6) RDPT of Pivot post for any crack (IC Sch) | Check ok | |
| | 7. Examine limit chain and links for wear, corrosion or damage | Check ok | |
| | 8. Inspect the safety slings, pins and R clips for wear and damage. | Check ok | |
| | 9. Check Aclathan/Vulkollon ring against any crack or shifting | Check | |
| | 10. Check gap in housing flange and traction link flange | Check | |
| | 11. Torque modified bolt M20 X65 by 260Nm (SMI 241) | चेक | |
| J | **BOGIE FRAME AND BRAKE GEAR** | चेक | |
| i | **BRAKE BLOCK** | चेक | |
| | 1. Inspect the brake blocks for wear. If it is below the minimum limit, replace the complete locomotive set brake block. | | |
| | 2. Check the brake block key security | | |
| | 3. Check the brake rigging for loose, damaged or worn out parts Rectify as required. | | |
| | 4. Examine the bogie frame, transom beam & damper base for signs of crack/damage | | |
| | 5. Check fitment of Palm Pull rod J Bracket, hanger lever and TM safety rope wire. | | |
| | 6. Inspect the rubber bumps for damage deterioration. Replace if required | | |
| | 7. Examine all bogies mounted equipment (Including earthing straps) for signs of damage and security of fastening | | |
| | 8. Inspect the brake rigging rotation points and component contact surfaces. Rectify any faults found. | | Check ok |
| | 9. Inspect the bogie body safety pin eyehole at the bogie frame for wear or cracks (IC Sch) | | 21/4/24 |
| | 10. Maintain the vertical & lateral clearances (Primary & secondary) as per TC-082 (IC Sch) | | 23/5/24 |
| | 11. Check the earthing shunts provided on bogie and body. | | |
| | Check intactness or axle box cover bolts **and sealing.** | | |
| | 12. Inspect brake gear for any worn out loose, damaged, hanging or dragging parts. Confirm intactness | | |
| | 13. Test for correct operation of the brake cylinder actuator unit. | | |
| | 14. Check the proper tightness of parking brake & service brake pneumatic pipes. | | |
| | 15. Check the external hitting marks & physical damage on the brake cylinder actuator unit. | | |
| | 16. Check the condition of parking brake cylinder bellows If damaged, replace them. | | |
| | 17. On parking brake cylinder pulling spindle & ring should be intact. | | |
| | 18. Check proper functioning of parking brake & service brake cylinder. | | |

**क्यूएजी जेई/एसएसई QAG-JE/SSE**

| Sr. No. | Detail of Work/ Inspection | Name Technician | Sign/Remark |
|---|---|---|---|
| | 19. Check the operation (manual releasing) of the parking brakes | Check | |
| | 20. Check the bogie isolating cock, should be in proper position. | Check | |
| | 21. Check the wear of scrubber blocks, replace if required (WAP5only) | Check | |
| | 22. Maintenance of Traction Motor support plate and bogie nose to Prevent crack/breakage of Traction Motor support plate (Holder for Traction motor suspension) should be done as per RDSO's SMI no. RDSO/2011/EL/SMI/0269 (Rev. '0') dtd, 18.05.11 | Check | |
| ii | **WHEEL SET AND AXLE BOX** | | |
| | 1. Examine all the wheels visually for following defects. | | |
| | a) Cracks: if any, requires replacement. | | |
| | b) Flats: Length 40mm or more requires re profiling. | | |
| | c) Cavities: Isolated cavities of length 15mm or more or multiple cavities of length 10mm or more and less than 50mm apart require re-profile/replacement. | | |
| | d) Metal builds up Height 1mm, or above and length 50mm or more requires re-profiling. For length less than 50mm., remove the build up with hand tools. | | |
| | e) Scoring and Grooving: Circumferentially around the wheel tread due to action of brake blocks require re-profile/replacement. | | |
| | 2. Measure and record the wheel diameters and tread profile. Refer RDSO's Tech. Circular no. ELRS/TC/0083 (Rev. "0"), (IC Sch) | Check ok | 27/5/24 |
| | 3. Disconnect the electrical cable from the earth return unit. Remove the earth return unit from the axle. Examine the bearings and rear sealing position for signs of grease leakage or deterioration. (IC Sch) | | |
| | 4. Check the axle for any cracks with the help of ultrasonic flaw detector. At the same time visual inspection should be done. (IC Sch Every 06 month) | | |
| | 5. Grease Axle box after 300000 KM Engine Run | | |
| | 6. Check the temperature of axle boxes. The axle boxes which are running warm or reported to have higher temperature in comparison with others shall be attended on priority by physically opening of axle box cover and outer angle ring. The suitable maintenance shall be done either in-situ or by dismantling the axle box. | | |
| iii | **GEAR CASE (WAP7/WAG9/WAG9H)** | | |
| | 1. Externally clean the gear case & its gauge glass. Inspect for any signs of oil leakage and any external hits/ damage. Verify integrity of gauge glass protective cover and clean oil sight glass for visibility. | | |
| | 2. Check the oil level in the gauge glass and top up if required. Ensure intactness of the drain plug along with its sealing. Check its tightness. | Check ok | |
| | 3. Check for any breakage of bolt and replace | | |
| | 4. Check for intactness of all fasteners of gear case | | 27/5/24 |
| | 5. Tighten all gear case fasteners at prescribed torque | | |
| | 6. Check the gear case oil visually and replace if found dirty or dust full level in the gear case and fill to the requirement | | |
| | 7. Check Gear case oil for metal contamination | | |
| | 8. Visually examine for fractures, the tyre profile for wear, flat, and skidding marks | | |

क्यूएजी जेई/एसएसई QAG-JE/SSE

| Sr. No. | Detail of Work/ Inspection | Name Technician | Sign/Remark |
|---|---|---|---|
| | 9. Check wheel root, flange and Tread wear & record Change all unserviceable Brake Blocks | | |
| | 10. Check condition and operation of Hand Brakes. Lubricate & adjust as necessary. | | |
| | 11. Adjust Brakes and lubricate tunion | | |
| | 12. Lubricate Brake Gearing, Slack Adjusters, Buffers, CB Coupler Assembly | | |
| | 13. Split the cotter pin, if new provided | | |
| | 14. Sch IC: a) Axle box old grease to remove and New grease to fill, old grease sample to be sent Lab for metal content check | | 25/5/24 |
| | b) Lubricate suspension roller tube bearing. | | |
| | c) Check Transition screw coupling long and short shackle bar thickness by Go-No Go gauge (39.5 mm snap 1/25 mm snap 8) | | |
| | d) Carry out DFT of Buffer Foundation and DPT/MPT of CBC Knuckle clevis &Yoke. Check size of Clevis, Knuckle, coupling by Go-No Go Gauge as specified. As per SMI-213/MIG-76 | | |

Check and put (✓) mark on the check box (CB) against each of the following items if the same has been checked. Write defects noticed and action takes in the Non-Schedule work table.

UNDER TRUCK [Inspect, Attend and Record]

| SN | Description | CB No. | Remarks |
|---|---|---|---|
| 1. | Measure and record buffer & rail guard height. | | |
| 2. | If skidded, fill Wheel Sledding Complain Report Farm, FRMB/22 or if Biased wheel wear, fill up Check last for Based Wheel Wear Cause Invest, FILM/24) | | |
| 3. | Proper Fitment of Brake Block and as Key | | |
| 4. | Rubbing of brake block with wheel in brake release position | | |
| 5. | Movement of brake pull rod in brake applied position | | |
| 6. | Brake Shoe and Brake hanger Pin/Bolts | | |
| 7. | Brake piston movement. | | |
| 8. | Support arrangement for brake cross tie bars | | |
| 9. | CBC Drooping. | | |
| 10. | Operation of both Transition screw couplings | | |
| 11. | Operation of CBC, Operating Rod and Coupler lock Pin | | |
| 12. | Operation of CBC Knuckles | | 25/5/24 |
| 13. | CBC retaining plate and support plate bolts | | |
| 14. | Side buffers and their mounting bolts. | | |
| 15. | Bolts of Cattle guards and back supports. | | |
| 16. | Fitment of Rail Guard | | |
| 17. | **Gear case mounting bolts and 'C' clamp bolts** | | |
| 18. | **Suspension bearing cap bolts and it's sealing.** | | |
| 19. | **Crack in TM Suspension taper roller bearing housing** | | |

**क्यूएजी जेई/एसएसई QAG-JE/SSE**

पश्चिम रेलवे
Western Railway

# SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI
## IA/IB/IC SCHEDULE OF 3PHASE ELECTRIC LOCOMOTIVE
## [COMPRESSOR &PNEUMATIC SYSTEM]

लोको संख्या Loco No 32649 क्लास Class शिड्यूल में लेने का दिनांकDate taken under Sch IA

| क्र.सं. SN | कार्यवाही कार्य निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (schedule) | कार्यवाही की Action Taken | टीसीएन का नाम Name of TCN | हस्ताक्षर/ टिप्पणी Sign/ Remarks |
|---|---|---|---|---|
| **A** | **मेन कंप्रेसर और बेबी कंप्रेसर Main Compressor and Baby Compressor** | | | |
| 1 | Check the pressure build up timing of each compressor independently in pre-inspection and take needful action during inspection. [ From 0 to 10 kg cm$^2$ i) With 1750LPM Comp-7 Minutes max. ii) With 1450LPM Comp- 8.5 Minutes max.] | CP-1→<br>CP-2→ | 6.5 - 6.8<br>6.0 - 7.0 | |
| 2 | Feel the vibrations of the compressor in pre-inspection and taken needful action during inspection. | checked ok | | 25/5/24 |
| 3 | Check for any external damage abnormal sound proper functioning of the compressor. | checked ok | | |
| 4 | Check the complete compressor support brackets for their healthiness. | checked ok | | |
| 5 | Visually examine the compressor mounting resilient blocks and take needful action | checked ok | | |
| 6 | Visually examine the various interconnecting pipelines and its support clamps of the compressors for their healthiness air leakage | checked ok | | |
| 7 | Cleaning of input air filter and the primary oil filter | checked ok | | |
| 8 | Clean the breather assembly | checked ok | | |
| 9 | Check and Drain the inter cooler after cooler assembly and also observeits discharge and take needful action | checked ok | | |
| 10 | Ensure healthiness intactness of various protection guards of compressor. | checked ok | | |
| 11 | Check the condition of the coupling between compressor & motor | checked ok | | |
| 12 | Visually check the compressor fan blade for their healthiness | checked ok | | |
| 13 | Check the oil level in the compressor and top up as perrequirement. | | | |
| 14 | Carry out the inspection of each type of compressor in line with recommendation of respective manufacturer. | checked ok | | 27/5/24 |
| 15 | Check compressor foundation base for crack (For under slung compressor) | checked ok | | |
| **B** | **Aux. Compressor (CPA)** | | | |
| 1 | Clean the CPA externally. Clean Suction Air filter. | checked ok | | |
| 2 | Check oil level and condition of oil and top up if required with the oil specified (Servo Press 150). | checked ok | | |
| 3 | Check tightness of all nut bolts, foundation bolts. | | | |
| 4 | Measure and record time & pressure of the auxiliary compressor | checked ok | | |
| **C** | **Sch IC**: i) CP all discharge valves to remove. | | | |
| | ii) CP all discharge valves refit by overhauled | | | |
| | **iii)** CP oil to be change. | Changed | | |
| | **iv)** Baby compressor/CPA oil to be change | | | |
| | **CPA** | — NA — | | |
| | **CP-1** | Changed | | |
| | **CP-2** | Changed | | |
| | **v)** Baby compressor/CPA safety valve (8.5 kg/cm$^2$) to be replace by overhauled | — NA — | | |

क्यूएजी जू. इंजी/सी.से.इंजी. के हस्ताक्षर Signature of QAG JE/SSE

| क्र.सं. SN | कार्यवाही कार्य /निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (schedule) | कार्यवाही की Action Taken | टीसीएन का नाम Name of TCN | हस्ताक्षर/ टिप्पणी Sign/ Remarks |
|---|---|---|---|---|
| D | AIR SUPPLY AND PNEUMATIC SYSTEM | | | |
| 1 | NRV and unloader valves fail due to ingress of Carbon which extracted from CP. So, one cycle dismantling, checking and cleaning of NRV and unloader to be ensured in all type of 3-phase locomotives. | Checked ok | | |
| E | AIR DRYER | | | |
| 1 | Inspect the air dryer for any damage and its mounting for any damage, crack in mounting plate, intactness of fasteners. Check for any hit mark on main reservoir and drain cock. Check for proper position of the air dryer angle cocks. | Checked ok | | |
| 2 | Check the integrity of the welding of the hanging support. | | | |
| 3 | Check humidity indicators for blue colour. White indications mean the air dryer is not operating effectively. Desiccant shall be changed, if needed. | Checked ok | | |
| 4 | Replace desiccant if found to be contaminated with either water or oil. | Checked ok | 25/5/24 | |
| 5 | Testing of air dryer for proper functioning to be done as per Technical Circular no. **RDSO/2011/EL/TC/0108, Rev.'0' dtd. 7.3.11** for testing of air dryers and its efficacy on electric locomotives. | Checked ok | | |
| F | **E-70 BRAKE SYSTEM (PNEUMATIC PANEL)** | | | |
| 1 | Visually inspect the pneumatic panel noting any sign of air leakages and rectify accordingly. | Checked ok | | |
| 2 | Clean the pneumatic panel externally. | | | |
| 3 | Check for correct position of the cut out isolating cock. | Checked ok | | |
| 4 | Check all the cable connections for looseness or deficiency of cleats. If any inspection section is to be informed **(IC Sch)** | | | |
| 5 | Ensure provisioning and proper functioning of RS gauge, Switch, lamp & other gauges on pneumatic panel. | Checked ok | | |
| 6 | Check the sealing of connectors, flexible pipe, regulator, air tank etc. | Checked ok | | |
| G | **BRAKE ISOLATING COCKS:** Check for correct operations of BP & FP angle cocks. Check the condition of BP/FP pipe, cock & isolating handle for any hit markor crack. Check correct condition of Bogie Isolating cock. | Checked ok | | |
| H | **BRAKE HOSES:** Examine the brake hoses at the ends of the locomotive to make sure that they are not chafed or damaged. Check couplingcocks for correct operation. | Checked ok | 25/5/24 | |
| I | **AIR RESERVOIR** | | | |
| 1 | Blow down and drain the reservoir until exhaust of complete water and oil. Check all drain cocks are closed on completion. | Checked ok | | |
| 2 | Check for any hit mark on main reservoir & drain cock | Checked ok | | |
| 3 | Ensure intactness / tightness of main reservoir foundation bolts. | Checked ok | | |
| 4 | Check the rubber gasket at the inspection cover of air reservoir for any damage/leakage. | Checked ok | | |
| 5 | Check the pressure regulator operates correctly | Checked ok | | |

क्यूएजी जू.इंजी/सी.से.इंजी.के हस्ताक्षर Signature of QAG JE/SSE

| क्र.सं. SN | कार्यवाही कार्य /निरीक्षण का विवरण / (शिड्यूल) Detail of Work/ Inspection (schedule) | कार्यवाही की Action Taken | टीसीएन का नाम Name of TCN | हस्ताक्षर/ टिप्पणी Sign/ Remarks |
|---|---|---|---|---|
| J | **THROTTLE VALVE (IC Sch):** Overhaul the throttle valve and change the rubber parts. | | | |
| K | **IN LINE FILTER (IC Sch):** Remove the filter element and clean thoroughly. | | | |
| L | **HORN & HORN SWITCH** | | | |
| 1 | Check condition of horn handle for any damage.Check for proper operation of horn. | checked ok | | |
| 2 | Check the condition of horn boots and replace if required | checked ok | | |
| 3 | Remove the horn & horn switch and Overhaul. Replace thediaphragm of horn. (**IC Sch**) | | | |
| 4 | Check all angle cocks for proper operation | checked ok | [illegible] 27/8/24 | |
| 5 | Ensure proper colour code of all the angle cocks. | checked ok | | |
| 6 | Check the tightness of the brake electronics control cards holding screws. | checked ok | | |
| 7 | Check all the pipes for leakage or damage. Check all the clamps for any damage and looseness. | checked ok | | |
| M | **वायुवीय प्रणाली PNEUMATIC SYSTEM** | | | |
| 1 | MR Safety valve, ADC, Panto throttle valve (Only for conventional Pantograph) replace by Overhaul in **IC Sch.** | | | |

**Remarks if any**

क्यूएजी जू .इंजी/सी.से.इंजी.के हस्ताक्षर **Signature of QAG JE/SSE**

पश्चिम रेलवे
Western Railway

**SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI**
**CHECK SHEET FOR 3PHASE HR PANTOGRAPH ASSEMBLY (IA/IB/IC)**

लोको संख्या Loco No 32649 क्लास Class IA शिड्यूल में लेने का दिनांक Date taken under Sch

SCHEDULE: ............. PANTOGRAPH SR.NO. /Make//Type: .................... LOCATION: ..............

| SN | Detail of Work/ Inspection (schedule) | Standard value | Action Taken | | Remarks |
|---|---|---|---|---|---|
| | | | PT-1 | PT-2 | |
| 1. | Ensure that wearing strips are without sharp edges and no gap between end strip and wearing strip. | Smooth and no gap | | | |
| 2. | Ensure availability of bolts, nuts, lock nuts, spring, washer, split pin etc. | All intact | Checked | Checked | |
| 3. | Check the rubber air pipe for any leakage | No leakage | Checked | Checked | |
| 4. | Check the rubber air bellow for any leakage/damage/rubbing | No leakage | | | |
| 5. | Check lower frame assembly for any crack and play | No crack | Checked | Checked | |
| 6. | Check upper frame assembly for any crack and breakage | No crack | Checked | Checked | |
| 7. | Check coupling rod assembly for crack and its fitting intactness. | No crack & intact | Checked | Checked | |
| 8. | Check diagonal rod for any crack both end side. (by DPT) | No crack | Checked | Checked | |
| 9. | Check plain bearing eye bush for freeness. | free | Checked | Checked | |
| 10. | Check ball bearing friction of Base frame, Upper frame and Coupling rod. | ok | Checked | Checked | |
| 11. | Check Cam wear on top of cable rope guide surface. | OK | Checked | Checked | |
| 12. | Check condition of all shunts (not more than 5% strand broken) | OK | Checked | Checked | |
| 13. | Check pantograph pan for any external hitting marks | No hitting marks | Checked | Checked | |
| 14. | Verify all welded joint for cracks. | OK | Checked | Checked | |
| 15. | Check the control box for any air leakage | No air leakage | Checked | Checked | |
| 16. | Check the condition and operation of hydraulic damper. | Normal | Checked | Checked | |
| 17. | Check Parallel guide bar for free function of Panto Pan Head | working | Checked | Checked | |
| 18. | Ensure that the thickness of wearing strips above the condemning size | New – 39 mm<br>Condemn – 26 mm | 8.10, 10.10 | 8.40 mm. | |
| 19. | Check the static contact force of pantograph at 6.4 $kg/cm^2$ MR pressure | 70 N /<br>7Kg Load Test | | | |
| 20. | Check the raising and lowering time of pantograph | 6 to 15 second | | | |
| 21. | Check the horizontality of panto pan with sprit level. | Level OK | | | |
| 22. | Check the function of auto drop device (ADD) | Working | ok<br>dummy | ok<br>dummy | |

Moti Lal

22.05.24

| SN | Detail of Work/ Inspection (schedule) | Standard value | Action Taken | | Remarks |
|---|---|---|---|---|---|
| | | | PT-1 | PT-2 | |
| 23. | Check the function of over reach detector (ORD) | **Working** | | | |
| 24. | Check air filter for air leakage or not draining. | **No leakage** | | | |
| 25. | Check the all pneumatic valve (3/2 valve, throttle valve and safety valve) for any leakage and proper function. | **No leakage** | | | |
| 26. | Check pressure regulator for any leakage and proper function. | **No leakage** | | | |
| 27. | Check the tightness of all bolts and thread locking compound to be provided in bearing base bolts. | **Provided** | | | |
| 28. | Examine the roofline conductors for defects/looseness at joints. | **No defect** | | | |
| 29. | Check the roof for any foreign material such as wire / clothpieces. | **No foreign material** | | | |
| 30. | Check tightness, cleanliness and wear-tear of shunt connection between Panto and Roof line. | **Tight, clean** | | | 27-05-24 |
| 31. | Check that the pan head locking pins springs are in place and secure. Check that the head moves freely on its mounting pins attached to the apex frame. If movement is not free, check that the head is not twisted, the side beam are not bent/twisted & mounting pins are not loose or bent. | **OK** | | | |
| 32. | Check tightness of the flexible shunt connecting screws. When a loose screw is found the shunt should be disconnected from the head &/or the pantograph frame, the mating surface cleaned & the shunt reconnected with new screws. Check all shunts for frayed strands and renew any shunt in which 5% or more of the strands are frayed. | **OK** | | | |
| 33. | **IC Sch:** Clean filter element, if found brownish then replace with new. | **Clear** | — | — | — |

**Remarks if any:**

**Signature and Name of Technician**

**Signature and Name of JE/SSE**

भारतीय रेलवे
Western Railway

# SABARMATI DIESELS, LOCO CARE CENTRE, SABARMATI

IA/IB/IC Schedule of 3 Phase Electric Locomotive [Heavy Section]

लोको संख्या Loco No 32649 क्लासClass WAG-9 HC IC शिड्यूल में लेनेका दिनांक Date taken under Sch

| Sr. No | Details of Work / Inspection (Schedule) | Action Taken | Name of TCN | Sign / Remarks |
|---|---|---|---|---|
| A | V C B | | | |
| 1 | Check insulator for crack, VCB roof bus bar, chips and flash marks. If damage in service, renew it | Checked | | |
| 2 | Clean insulator, other parts that are dirty with soft, clean & dry cloth. (Do not use cleaning products based on fluoridated or chlorate compounds or sodium meta silicate), apply Silicon Spray. | Checked | Harl | |
| 3 | Check contact spring (Earthing contact) | Checked | Dinesh | |
| 4 | Check moving contact lever of auxiliary switch and tension spring & its mounting bolts for breakage | Checked | | |
| 5 | Check the cables & lugs for broken cables, lugs looseness | Checked | | |
| 6 | Check the main bushing condition, its connection and foundation bolts and clamps. | Checked | | |
| 7 | Check the tightness of HV Connections (Torque 70Nm) | Checked | | |
| 8 | IC Sch:<br>a) Check for damage to connection of the earthing isolator, Cleaning & greasing, if necessary. | NA | | |
| | b) Check the torque of VCB fixing screw (Torque 70Nm) | NA | | |
| | c) Remove the bottom cover and check that the control air and electrical connection are tightened & Ensure that there is no leakage from pressure regulator | NA | | |
| B | EARTHING SWITCH (HOM) | | | |
| | a) Check the arm of earthing switch for any sign for any defect, play, crack etc, copper braid is undamaged and properly connected | | | |
| | b) Clean and lubricate the copper braid | | | |
| C | PRESSURE SWITCH (IC Sch) | | | |
| | Check the tightness of pneumatic connection | | | |
| D | AUXILIARY SWITCH (IC Sch) | | | |
| | a) Check drive screw for breakage or other damage | NA | | |
| | b) Check the parts of control rod and stay bolt for any deformation | | | |
| | c) Check the cables and lugs for broken cables, lugs looseness. | | | |
| | d) Securing of auxiliary blocks and support (Tightening torque-1.2Nm) | | | |
| | e) Check the working order of each auxiliary contacts | | | |
| E | EARTHING CONNECTION. | | | |
| | a) Check the earthing connection tightening torque 70Nm. | | | |
| | b) Connection of earthing system in 3-phase locomotives type WAP5, WAP7 & WAG9 (As per RDSO/2007/EL/SMI/248 Rev.'0' dated 22.11.2007) | | | |

Signature of QAG JE/SSE

| Sr. No | Details of Work / Inspection (Schedule) | | | | Action Taken | Name of TCN | Sign / Remarks |
|---|---|---|---|---|---|---|---|
| | c) Check Earthing shunt of Bogie to Body | | | | | | |
| | Axle box 1-3 | | Axle box 2-4 | | | | |
| | Axle box 5-7 | | Axle box 6-8 | | | | |
| | d) Check Earthing shunt of Axle to Bogie | | | | | | |
| | Axle box 2 | | Axle box 3 | | | | |
| | Axle box 6 | | Axle box 7 | | | | |

| Sr. No | Details of Work / Inspection (Schedule) | Action Taken | Name of TCN | Sign / Remarks |
|---|---|---|---|---|
| F | **MAIN TRANSFORMER** | | | |
| 1 | Read off the oil level on the gauge situated on the conservator. Top up the oil as necessary and check for any signs of leakage. Clean the gauge with dry cloth | OK | | |
| 2 | Inspect the colour of the silica gel. If it is pink, replace with blue silica gel | OK | | |
| 3 | Examine the flanges of the pipe couplings and flexible hose that link the transformer and conservator | OK | | |
| 4 | Check the availability of hosepipe over the oil pipe compensator and check stoochi coupling pipes for proper layout fitment. | OK | | |
| 5 | Check the main Transformer and its protection cover for any damage, crack & oil leakage. Also refer instructions contained in **RDSO's letter no. EL/3.2.1/3-Ph / TC-0076** for arresting oil leakages cases" | OK | | 22/07/24 |
| 6 | Check the deformity of Transformer drain cock protection cover guard if deformed replace it | OK | | |
| 7 | Perform the sample test on transformer oil. Check specific value of BDV, DGA, moisture and acidity and condition monitoring of loco transformer by Dissolve Gas Analysis. **(RDSO/SMI/138 & OEM Doc. Dt. 27th Nov. 1995)** | OK | | |
| 8 | Check visually the condition of transformer foundation bolt and Nylon lock nuts for proper locking. | OK | | |
| 9 | Check the earthing cable. Clean and visually inspect the insulator condition and Electrical connection of transformer and converter for crack & flashed mark and ensure red marking | OK | | |
| 10 | Visually inspect for leakage, damage, burning/overheat signs and all fixing clamps condition of oil cooling metallic pipes, Prismatic level gauge, TFP bushing, HV bushing, Hose pipe etc. | OK | | |

**Signature of QAG JE/SSE**